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# The calibration of stochastic local-volatility models: An inverse problem perspective

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ABSTRACT

We tackle the calibration of the Stochastic Local-Volatility (SLV) model. This is the class of financial models that combines the local volatility and stochastic volatility features and has been subject of the attention by many researchers and practitioners recently. The corresponding inverse problem consists in finding certain (functional) coefficients in a class of parabolic partial differential equations from observed values of the solutions. More precisely, given a calibrated local volatility surface and a choice of stochastic volatility parameters, we calibrate the corresponding leverage function. Our approach makes use of regularization techniques from inverse-problem theory, respecting the integrity of the data and thus avoiding data interpolation. The result is a stable and efficient algorithm which is resilient to instabilities in the regions of low probability density of the spot price and of the instantaneous variance. We substantiate our claims with numerical experiments using synthetic and real data.

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## 1. Introduction

The search for parsimonious models that would capture the market-observed smile behavior in the Implied Volatility Surface (IVS) is still one of the main research topics in Quantitative Finance. It falls in the cross-roads of mathematical modeling, optimization, and numerical analysis of inverse problems for partial differential equations. Among the different models that have been introduced, perhaps the two most important attempts are the Stochastic Volatility (SV) models, e.g. [1] and [2], and the Local Volatility (LV) model of [3]. While SV models capture crucial stylized facts of the volatility dynamics, they cannot perfectly calibrate the IVS, especially for short maturities. On the other hand, the LV model was constructed to fit any arbitrage-free IVS. However, it has poor dynamical properties, see [4]. A very important issue when considering these models is their calibration to the market-observed IVS; we refer the reader to [5–9] and references therein for different calibration methods of SV and LV models, separately.

The Stochastic Local-Volatility (SLV) model is able to combine the best aspects of each one of such model classes, see [10–12]. In the present article we shall present a stable and effective method to calibrate the SLV model that consists of adapting the method proposed in [13] and [14] to the SLV framework.

Although the calibration of SV and LV models, separately, has been extensively discussed in the literature, to the best of our knowledge, there is a lack of methods of calibration of SLV models. Indeed, to address this issue two avenues can be pursued. At one direction, we have Monte Carlo methods, [10,15]. At another direction, one might rely on numerical

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methods for partial differential equations (PDE) as in [12]. Since the method we propose here is also based on PDEs, we will use as *benchmark* the method presented [12]. Additionally, [16] uses the same calibration idea as in this benchmark method, but considers an adjoint method to solve the related Fokker–Planck equation for the transition probability density, see Section 3.1.

In order to illustrate our method, we consider two numerical exercises. One uses synthetic data generated from a known SLV model and the other uses real option data from an FX market. These examples corroborate the theoretical conclusions of the comparison of the benchmark and our proposed method. In fact, we verify that our method is more robust against noise and more resilient to instabilities.

The paper is organized as follows. In Section 2, we briefly describe the SLV model. The benchmark and proposed calibration procedures are outlined in Section 3. Finally, in Section 4, we test our method with synthetic and real FX data. We add the Appendix with the specification of the Alternating Direction Implicit algorithm (ADI) for the numerical solution of the PDEs considered here.

## 2. Model description

The Stochastic Local-Volatility (SLV) model assumes that, under a risk-neutral measure in a given probability space, the spot price satisfies

$$\begin{cases} dS_t = (r-d)S_t dt + \sqrt{V_t}L(t,S_t)S_t dW_t^S, \\ dV_t = \kappa(m-V_t)dt + \xi\sqrt{V_t}dW_t^V, \\ dW_t^S dW_t^V = \rho dt, \end{cases} \tag{2.1}$$

where $W^S$ and $W^V$ are correlated standard Brownian motions under this risk-neutral measure.

The rates $r$ and $d$ are the risk-free interest rate and the dividend rate, respectively. In this version of the SLV model, we assume that the stochastic part of the volatility is following the Heston model, [1]. The parameters $\kappa$, $m$, $\xi$ and $\rho$ have the same interpretation as in the pure SV model. Moreover, notice that this SLV model simplifies to the Heston model when $L \equiv 1$. With respect to our proposed calibration procedure, the choice of the SV dynamics could be easily modified. For example, we could have considered the SABR model of [17] or the Inverse Gamma model of [18]. Additionally, it is fairly easy to extend the method presented here to deal with time dependent interest and dividend rates, as we consider in our numerical examples. However, for cleaner exposition, we will consider constant rates.

The function $L$ is called the *leverage function* and it plays a very important role in the model above. It is the ingredient that allows the model to perfectly calibrate the IVS seen in the market. In order to achieve this goal, the function $L$ must satisfy (see [19])

$$\sigma_{loc}^2(t,S) = \mathbb{E}[V_t L^2(t,S_t) \mid S_t = S] = L^2(t,S)\mathbb{E}[V_t \mid S_t = S], \tag{2.2}$$

where $\sigma_{loc}$ is the local volatility function calibrated to the market. We define then

$$\Sigma(t,S) = \mathbb{E}[V_t \mid S_t = S]. \tag{2.3}$$

It is important to notice that Eq. (2.2) is an implicit equation for $L$, since it is needed for the computation of $\Sigma(t,S)$.

Note that the parameters of the SV part of the model may be (almost) freely chosen. Given a reasonable choice of parameters, choosing $L$ to satisfy Eq. (2.2) allows the model to fit any arbitrage-free IVS. The adjectives *almost* and *reasonable* used here refer to the fact that the SDE (2.1) using formula (2.2) for $L$ might not have a solution for certain choices of parameters, see Remark 3.1.

## 3. Calibration

In this section we shall discuss two different PDE techniques that can be applied to calibrate an SLV model, namely the *benchmark* and our proposed methods. For both, we will assume that the local volatility surface and the SV parameters of the model have been already computed.

Notice now that we can rewrite Eq. (2.3) as

$$\Sigma(t,S) = \mathbb{E}[V_t \mid S_t = S] = \frac{\int_0^{+\infty} Vp(t,S,V)dV}{\int_0^{+\infty} p(t,S,V)dV}, \tag{3.1}$$

where $p(t,\cdot,\cdot)$ is the joint density probability $(S_t, V_t)$ and solves the Fokker–Planck PDE:

$$\begin{aligned} &\frac{\partial p}{\partial t} + \frac{\partial}{\partial S}((r-d)Sp) + \frac{\partial}{\partial V}(\kappa(m-V)p) - \frac{1}{2}\frac{\partial^2}{\partial S^2}(VL^2(t,S)S^2p) \\ &\quad - \frac{1}{2}\frac{\partial^2}{\partial V^2}(\xi^2 Vp) - \frac{\partial^2}{\partial S \partial V}(\rho\xi VL(t,S)Sp) = 0, \end{aligned} \tag{3.2}$$

with initial condition $p(0,S,V) = \delta(S-S_0)\delta(V-V_0)$, i.e. the Dirac mass at $(S_0, V_0)$.

**Remark 3.1** (*Existence of Solution for SDE (2.1)*)**.** Using Eq. (3.1), we may rewrite the SDE (2.1) as

$$\begin{cases} dS_t = (r-d)S_t dt + \sqrt{V_t}\sigma_L(t,S_t)\sqrt{\dfrac{\int_0^{+\infty} p(t,S_t,V)dV}{\int_0^{+\infty} Vp(t,S_t,V)dV}}S_t dW_t^S, \\ dV_t = \kappa(m-V_t)dt + \xi\sqrt{V_t}dW_t^V, \\ dW_t^S dW_t^V = \rho dt. \end{cases} \tag{3.3}$$

This is called a *McKean–Vlasov SDE*, since the diffusion coefficient depends on the law of $(S, V)$. The existence of solutions of this SDE is a very challenging problem, and is outside the scope of this paper. For a discussion of this topic, see [10] and [20]. For our work here, we will assume that the SDE has a unique strong solution.

**Remark 3.2** (*Mixing Fraction*)**.** Additional parameters could be considered in order to calibrate some exotic derivatives (e.g. Barrier or Asian options). In particular, given some fixed vol-of-vol, $\xi$, and correlation, $\rho$, one could define

$$\xi_\lambda = \lambda\xi \text{ and } \rho_\lambda = \lambda\rho,$$

for $\lambda \in [0, 1]$. This parameter is usually called *mixing fraction*, as it mixes the stochastic and local aspects of the volatility. Notice that $\lambda = 0$ implies a pure LV model. Moreover, the parameters $\xi$ and $\rho$ could be taken as the calibrated parameters of a pure SV model. The goal is then to choose $\lambda$ in order to calibrate a given exotic derivative price. For instance, if we choose a *down-and-out barrier call option* with barrier $B$ and strike $K > B$, we could numerically solve the following PDE

$$\begin{aligned} \frac{\partial P}{\partial t} + (r-d)S\frac{\partial P}{\partial S} + \frac{1}{2}vL^2(t,S)S^2\frac{\partial^2 P}{\partial S^2} + \kappa(m-v)\frac{\partial P}{\partial v} + \frac{1}{2}\lambda^2\xi^2 v\frac{\partial^2 P}{\partial v^2} \\ + \lambda^2\xi\rho vL(t,S)S\frac{\partial^2 P}{\partial S\partial V} - rP = 0, \end{aligned} \tag{3.4}$$

for $S \in [B, +\infty)$, with $P(t, B, V) = 0$ and final condition $P(T, S, V) = (S-K)^+$. It is straightforward to consider a time-dependent $\lambda$.

**Remark 3.3** (*Monte Carlo Methods*)**.** For multi-factor SV models, both methods described in this paper require a high-dimensional PDE solver to numerically deal with the Fokker–Planck equation and therefore suffer from the curse of dimensionality. In [15], the author considered the case in which the variance process is driven by two independent Brownian motions. By constructing a map between the prices driven by SV only and SLV, an algorithm is proposed to calculate the conditional expectation of the variance in (2.3) by using Monte Carlo methods. By the equation for the leverage function (2.2), the algorithm estimates $L$ at each time step. In [10], the authors applied McKean's particle method, and developed an algorithm for models where the short-term rate and the volatility are modeled as diffusions.

### 3.1. Numerical aspects

There are some common numerical aspects for both calibration procedures, and we will state them here. Firstly, since the methods considered here are based on finite difference methods for PDEs, we will consider discrete meshes for time $(t)$, spot price $(S)$ and spot volatility $(V)$. A (uniform) mesh for a variable $y$ depends on a choice for a finite lower bound $y_{\min}$, a finite upper bound $y_{\max}$ and a step size $\Delta y$. It is assumed that $N_y = (y_{\max} - y_{\min})/\Delta y \in \mathbb{N}$. The mesh for $y$ is then

$$y_i = y_{\min} + i\Delta y, \text{ for } i = 0, \ldots, N_y.$$

In our case, we will assume that $t_{\min} = S_{\min} = V_{\min} = 0$. We will use the sub-index $n$ for $t$, $i$ for $S$ and $j$ for $V$. One could surely use non-uniform meshes, but we will present the results here with uniform meshes for clearer illustration.

#### *Numerical methods for the Fokker–Planck PDE*

The Fokker–Planck PDE, shown in Eq. (3.2), will have to be numerically solved given the parameters of the SV model and a fixed leverage function $L$. That is, discretizing the Fokker–Planck PDE with any chosen method, we will compute an approximation for $p(t_n, S_i, V_j)$. A sensible choice for the numerical method is an Alternating Direction Implicit (ADI) method, see [21]. Namely, in our numerical example, we consider the Douglas scheme, which was proposed in [22]. Moreover, the choice of boundary conditions for the numerical method is also very important. We have chosen the zero flux condition, see for instance [23]. For more details, see the Appendix. Note that, by using an ADI method to solve the Fokker–Planck equation, the approximation of $p$ at time $t_{n+1}$ depends on $L$ at time $t_n$ and $t_{n+1}$.

More precisely, both calibration algorithms presented here compute the leverage function sequentially in time. So, let $L_n = \{L_{n,i}\}_{i=0}^{N_S}$ be the approximation of the leverage function at time $t_n$ in the mesh for $S$ and $\ell = \{\ell_i\}_{i=0}^{N_S}$ be the proposed value of the leverage function at time $t_{n+1}$ in the same mesh. Then the density function $p(t_{n+1}, \cdot, \cdot)$ is approximated by the ADI method discussed above setting $L(t_n, S_i) = L_{n,i}$ and $L(t_{n+1}, S_i) = \ell_i$. We denote this approximation by $p_{n+1,i,j}$. In other

words, we define the one-step finite difference operator $\mathcal{G}_{FP}$ that associates the given values for the leverage function, $L_n$ and $\ell$, and the density, $p_n = \{p_{n,i,j}\}_{i,j=0}^{N_S,N_V}$, to the corresponding approximation of the density at time $t_{n+1}$:

$$\mathcal{G}_{FP}(p_n, L_n, \ell) := p_{n+1}. \tag{3.5}$$

As we will notice, the benchmark method assumes that this approximation only depends on $L$ at time $t_n$, i.e. it sets $\ell = L_n$. In a different direction, one could consider adjoint methods to numerically solve the Fokker–Planck PDE as in [16].

*Approximation of the initial condition*

The initial condition for our Fokker–Planck PDE is not well-behaved; it is a Dirac mass at the point $(S_0, V_0)$. In order to avoid numerical issues arising from this lack of smoothness, we consider a smooth approximation of this Dirac mass. Specifically, we use a bivariate normal distribution with small variances to represent the initial density:

$$p(0, S, V) = \frac{1}{2\pi\sigma_S\sigma_V} \exp\left\{-\frac{1}{2\sigma_S^2}(S - S_0)^2 - \frac{1}{2\sigma_V^2}(V - V_0)^2\right\}. \tag{3.6}$$

In our numerical experiment, we have used $\sigma_S = \sigma_V = 10^{-3}$. See, for instance, [12] for details.

*Numerical computation of the local volatility*

The calibration of local volatility surfaces is an important inverse problem in Quantitative Finance. In [3], the author has proposed the local volatility model, in which the European options prices satisfy the PDE of the form

$$\frac{\partial C}{\partial T} + (r - d)K\frac{\partial C}{\partial K} - \frac{1}{2}\sigma_{loc}^2(T, K)K^2\frac{\partial^2 C}{\partial K^2} + dC = 0, \quad T > 0, K > 0, \tag{3.7}$$

with initial and boundary conditions given by

$$\begin{aligned} C(0, K) &= (S_0 - K)^+, \\ \lim_{K\to\infty} C(T, K) &= 0, \\ \lim_{K\to 0} C(T, K) &= S_0, \end{aligned} \tag{3.8}$$

where $T$ is expiration date, $K$ is the strike price, and $C = C(T, K)$ is the value of the European call option with expiration date $T$ and strike price $K$. The inverse problem of the local volatility model is that, given the options prices $\{C(T, K)\}_{T,K}$, we want to find a plausible local volatility surface, $\{\sigma_{loc}(T, K)\}_{T,K}$, which can explain these options prices. Two of the challenges of this inverse problem are the ill-posedness, [24], and the scarceness of the data of options prices, [5]. To solve an ill-posed inverse problem, one popular method is to use the Tikhonov regularization [24–28]. We will briefly introduce this regularization method in Section 3.3. To solve the problem of data scarcity, one possibility is to interpolate/extrapolate the options prices data to all the locations of the mesh, [29]. In this paper, however, we apply the method discussed in [5,30], where we use a restriction operator to map the grid locations of the estimated options prices to those of real data.

### 3.2. Benchmark method

In this section, we will describe the method proposed in [31] and further developed in [12], which is our benchmark method. From Eq. (2.2), we have

$$L(t, S) = \frac{\sigma_{loc}(t, S)}{\sqrt{\Sigma(t, S)}} = \sigma_{loc}(t, S)\sqrt{\frac{\int_0^{+\infty} p(t, S, V)dV}{\int_0^{+\infty} Vp(t, S, V)dV}}.$$

The *benchmark* calibration procedure is based on the equation above. As we have previously mentioned, this is an implicit equation for $L$, since $p$ depends on it. More precisely, the leverage function is initialized at a mesh point $S_i$ as

$$L_{0,i}^B = \frac{\sigma_{loc}(0, S_i)}{\sqrt{V_0}}, \tag{3.9}$$

We are using the superscript $B$ to denote that this is the leverage function computed by the benchmark method. Assuming we have computed $L^B$ at time $t_n$, we use the numerical method discussed in Section 3.1 to solve the Fokker–Planck Equation (3.2) from $t_n$ to $t_{n+1}$:

$$p_{n+1}^B = \mathcal{G}_{FP}(p_n, L_n^B, L_n^B),$$

where we are setting the leverage function at time $t_{n+1}$ equals the calibrated leverage function at time $t_n$. Finally, we update $L^B_{n+1}$ using

$$L^B_{n+1,i} := \sigma_{loc}(t_{n+1}, S_i)\sqrt{\frac{\sum_{j=0}^{N_V} p^B_{n+1,i,j}}{\sum_{j=0}^{N_V} V_j p^B_{n+1,i,j}}}, \tag{3.10}$$

and repeat the procedure above varying $n$ from 0 to $N_t - 1$.

**Algorithm 1** Benchmark Algorithm of [31]

1: Set the initial condition of $p_{0,i,j}$ and $L^B_{0,i}$ using Equations (3.6) and (3.9), respectively.
2: **for** $n = 0, 1, 2, \ldots, N_t - 1$ **do**
3: Set $L(t, S_i) = L^B_{n,i}$, for $t \in [t_n, t_{n+1}]$.
4: Solve the finite difference problem in (3.5) for $\{p_{n+1,i,j}\}_{i,j=0}^{N_S,N_V}$.
5: Update $L^B_{n+1,i}$ with Equation (3.10).
6: **end for**
7: **return** $L^B_{n,i}$ for $n = 0, \ldots, N_t$ and $i = 0, \ldots, N_S$.

### 3.3. Proposed method

The problem under consideration is an ill-posed one. One possible avenue to tackle this difficulty is Tikhonov regularization [32]. The latter transforms solving $F(x) = y$, into a minimization of the form

$$\arg\min_x \|F(x) - y\|^2 + \alpha |||x - x_0|||^2,$$

where $\|\cdot\|$ and $|||\cdot|||$ are two norms, and $x_0$ incorporates the *a priori* information that will allow the regularization of the problem. By changing the scale factor $\alpha$ of the norm $|||\cdot|||$, one would put more (or less) emphasis on such *a priori* information. The optimal choice of $\alpha$ is the subject of intense investigation. Among the most well-known methods one can cite the discrepancy principle and the $L$-curve method, see [32]. Further, developments led to the use of other metrics (or more generally functionals instead of norms), see [33] and references therein.

Let $L_n = \{L_{n,i}\}_{i=0}^{N_S}$ be the leverage function at time $t_n$ and in the mesh of the spot price $S$ computed using our proposed method described below. Moreover, consider the possible choice of the leverage function at time $t_{n+1}$ to be given by $\ell = \{\ell_i\}_{i=0}^{N_S}$. Then the density function $p(t_{n+1}, \cdot, \cdot)$ is approximated by the numerical method discussed in Section 3.1. We denote this approximation by $p_{n+1}$:

$$p_{n+1} = \mathcal{G}_{FP}(p_n, L_n, \ell).$$

The initialization $L_0 = \{L_{0,i}\}_{i=0}^{N_S}$ will be discussed in Remark 3.4.

Let $\mathcal{G}_{LV}$ be the operator mapping this possible choice of leverage function at $t_{n+1}$ to the local volatility function at time $t_{n+1}$ following Eq. (2.2). Thus

$$\mathcal{G}_{LV}(p_{n+1}, \ell) := \left\{ \ell_i \sqrt{\frac{\sum_{j=0}^{N_V} V_j p_{n+1,i,j}}{\sum_{j=0}^{N_V} p_{n+1,i,j}}} \right\}_{i=0}^{N_S}. \tag{3.11}$$

Notice that

$$\mathcal{G}_{LV}(p_{n+1}, \ell) = \mathcal{G}_{LV}(\mathcal{G}_{FP}(p_n, L_n, \ell), \ell).$$

We shall now define

$$\mathcal{G}(p_n, L_n, \ell) = \mathcal{G}_{LV}(\mathcal{G}_{FP}(p_n, L_n, \ell), \ell).$$

In other words, $\mathcal{G}$ is the operator that takes the computed value of the leverage function at time $t_n$, $L_n = \{L_{n,i}\}_{i=0}^{N_S}$, the proposed value for the leverage function at time $t_{n+1}$, $\ell = \{\ell_i\}_{i=0}^{N_S}$, and the computed value of the probability density at time $t_n$, $p_n = \{p_{n,i,j}\}_{i,j=0}^{N_S,N_V}$, to the local volatility at time $t_{n+1}$. To obtain the surface of the leverage function, we would have to solve an ill-posed inverse problem. We propose to do that by solving the following Tikhonov-type optimization problem for $n = 0, 1, \ldots, N_t - 1$.

$$\begin{aligned} L_{n+1} :=& \arg\min_{\ell} \Big\{ \|\sigma_{loc}(t_{n+1}, \cdot) - \mathcal{G}(p_n, L_n, \ell)\|^2_{\Gamma^{-1}} \\ &+ \alpha_1 \|\ell - L_n\|^2_{D_0^{-1}} + \alpha_2 \|R_S \ell\|^2_{D_S^{-1}} \Big\}, \end{aligned} \tag{3.12}$$

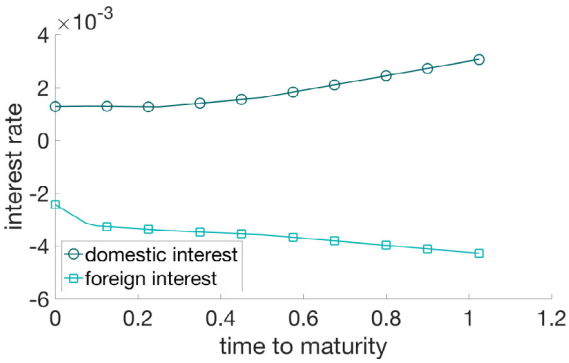



**Fig. 1.** Domestic and foreign interest rates.

where $\Gamma, D_0$ and $D_S$ are chosen as symmetric positive definite covariance matrices. Here the norm is defined as $\|x\|_C = \sqrt{x^T C x}$ and $R_S$ is the matrix representing the finite-difference approximation of the linear operator $\partial_S$.

**Remark 3.4.** The initial value $\{L_{0,i}\}_{i=0}^{N_S}$ is chosen by solving the minimization (3.12) with $L_{-1,i} = c$, for all $i = 0, \ldots, N_S$, for some chosen constant $c$. A sensible choice would be $c = 1$. The reason for such prior information is the following: since we are using the stochastic volatility parameters calibrated to the observed implied volatility, it is expected the conditional expectation in Eq. (2.2) to be close to the local volatility calibrated to the market. Hence, the leverage function would be close to 1. The discrepancy from this value depends on the distance between the pure SV model implied volatility and the observed one.

**Remark 3.5** (*Improvements Over the Benchmark Method*)**.** The main improvements are that our proposed method provides the regularized solution of the problem at each step and the numerical method to solve the Fokker–Planck PDE does not set the leverage function equal at $t_n$ and $t_{n+1}$. Furthermore, for the initial step of $L$, it yields the regularized best fit to data according to (3.12). These are not achieved in the benchmark method.

**Algorithm 2** Proposed Algorithm

1: Set the initial condition of $\{p_{0,i,j}\}_{i,j=0}^{N_S,N_V}$ using Equation (3.6) and set $\{L_{-1,i}\}_{i=0}^{N_S}$ to a chosen constant.
2: **for** $n = 0, 1, 2, \ldots, N_t - 1$ **do**
3: Solve the minimization problem (3.12) for $\{L_{n+1,i}\}_{i=0}^{N_S}$.
4: Solve the finite difference problem in (3.5) for $\{p_{n+1,i,j}\}_{i,j=0}^{N_S,N_V}$.
5: **end for**
6: **return** $L_{n,i}$ for $n = 0, \ldots, N_t$ and $i = 0, \ldots, N_S$.

**Remark 3.6.** In our numerical experiments, we have chosen $\alpha_1 = 0$ to limit the impact of our prior information on the leverage function (which is introduced in our algorithm through the value of $L_{-1,i}$). Including an arbitrary positive value for $\alpha_1$ gives the methodology more freedom but requires the choice of both parameters. The choice of $\alpha_1$ and $\alpha_2$ can be done for example by means of the techniques suggested in [34]. After setting $\alpha_1 = 0$, we carefully studied the impact of $\alpha_2$ using the $L$-curve method and then chose it accordingly.

## 4. Numerical example

We will now compare the methods described in Sections 3.2 and 3.3 within synthetic and real data examples. The interest rates used in both examples are given in Fig. 1.

In Section 4.3, we will show the numerical results obtained in each of the examples below. We postpone the discussion and comparison to the end of this section.

### 4.1. Synthetic data

In this synthetic data example, we suppose the *ground truth* leverage function is given by

$$L(t, x) := 1.1^{4\cos(2\pi x t)}, \tag{4.1}$$

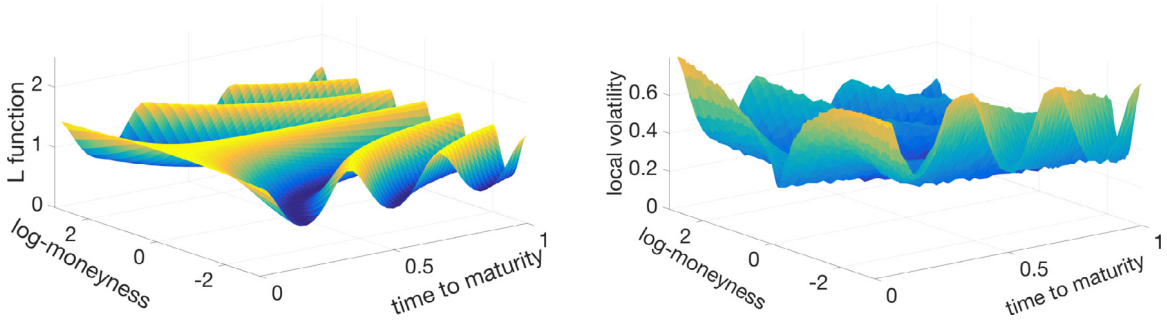



**Fig. 2.** The ground truth leverage function (left) and the local volatility surface $\sigma_{loc}(t,x)^{\eta}$ (right).

**Table 1**
Mesh parameters for the synthetic data example.

| Variable | Lower bound | Upper bound | Fine mesh | Coarser mesh |
|---|---|---|---|---|
| Time | 0 | 1 | 0.001 | 0.025 |
| Log-moneyness | −3 | 3 | 0.025 | 0.05 |
| Volatility | 0 | 1 | 0.005 | 0.01 |

**Table 2**
SV parameters for the synthetic data example.

| Parameter | Value |
|---|---|
| $V_0$ | 0.04 |
| $\kappa$ | 2 |
| $m$ | 0.04 |
| $\xi$ | 0.25 |
| $\rho$ | −0.5 |

where the log-moneyness $x \in [-3,3]$ and $t \in [0,1]$, as shown in Fig. 2. We calculate $\Sigma(t,x)$ and local volatility surface $\sigma_{loc}$ based on this given $L$ in the fine mesh. The details of the mesh for maturity, log-moneyness and volatility are given in Table 1. We then add a relative noise to the local volatility surface

$$\sigma_{loc}(t,x)^{\eta} := \sigma_{loc}(t,x)(1+0.01\eta_{t,x}) \tag{4.2}$$

where $\eta_{t,x}$ are independent draws from the standard normal distribution $\mathcal{N}(0,1)$. Numerical simulations of the data are commonly used in testing algorithms for the solution of inverse problems. Using the same numerical model and parameters tends to give over-optimistic results. This is known as the so-called *inverse crime* [35]. In order to avoid such problem we simulated the data in a fine mesh. Then we sampled the data to a coarser mesh, added noise and solved the inverse problem in the coarser mesh. The mesh description is given in Table 1. The use of such a mesh guarantees a different numerical model although ideally one should perhaps change also the solution technique.

Fig. 2 presents the noisy local volatility surface. The parameters of the SV part of the model are given in Table 2. The Tikhonov parameters are chosen to be $\alpha_1 = 0$ and $\alpha_2 = 10^{-2}$, see Eq. (3.12), and the initial constant is $c = 1$, see Remark 3.4. In order to further understand the dependence of the algorithm with respect to the parameter $\alpha_2$, we also tested $\alpha_2 = 10^{-1}$ and $\alpha_2 = 10^{-3}$. They generated the relative errors 1.61% and 1.44%, respectively, cf. Table 5. Therefore, one notices that choosing a large $\alpha_2$ might create a leverage function that is too smooth and does not fit the observed data very well increasing the relative error. On the other hand, choosing a value for $\alpha_2$ that is too small might over-fit the observed data without achieving a significant reduction in the relative error, cf. Table 5.

### 4.2. Real data

In this section we present an example using real data. We chose FX options on EURUSD on March 18th, 2015. They include the typical 25 liquid option contracts, with 5 maturities (1 W, 1M, 3M, 6M, 1Y) and 5 strikes (related to 10 and 25 Call and Put Delta and to ATM) per maturity (see Fig. 3). The spot value was 1.0864. The parameters of the Heston model are calibrated to this data set and given in Table 4.

These parameters were required to satisfy the Feller condition. This translates into more realistic dynamics for the volatility, since it prevents the process to reach the zero volatility boundary. The domestic and foreign interest rates are the same as in Fig. 1. The discretization parameters are given in Table 3. In Fig. 3, we show the estimated local volatility surface from option prices. For the description of the methods that we used to calibrate the local volatility surface, see Section 3.1. The Tikhonov parameters are chosen to be $\alpha_1 = 0$ and $\alpha_2 = 10^{-4}$, see Eq. (3.12), and the initial constant is $c = 1.0$, see

**Table 3**
Mesh parameters for the real data example.

| Variable | Lower bound | Upper bound | Regular mesh |
|---|---|---|---|
| Time | 0 | 1 | 0.01 |
| Log-moneyness | −0.5 | 0.5 | 0.02 |
| Volatility | 0 | 1 | 0.01 |

**Table 4**
SV parameters calibrated to real data.

| Parameter | Value |
|---|---|
| $V_0$ | 0.013 |
| $\kappa$ | 1.025 |
| $m$ | 0.013 |
| $\xi$ | 0.161 |
| $\rho$ | −0.626 |

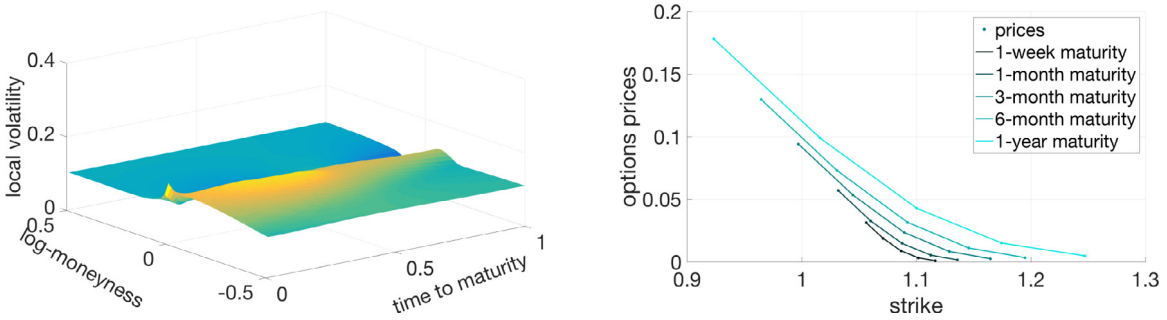



**Fig. 3.** EUR-USD local volatility surface and options prices on March 18th, 2015.

Remark 3.4. We recall that in the synthetic example, the noise was added to local volatility creating a rather rough surface. On the other hand, in the real data examples, the local volatility surface calibrated by methods proposed in [5] come with some smoothness. Hence, in the real data example, over-fitting has already been dealt with by the regularization process for the calibration of the local volatility surface. Thus we may choose a small value for $\alpha_2$ . In both examples, we choose $\alpha_1 = 0$ to limit the effect of the prior information on the leverage function.

### 4.3. Discussion of the numerical results

The computer used for all the examples is an Intel Core i7 with a CPU of 2.9 GHz and 8 GB of RAM. The software used was Matlab. The running time is presented in Table 5. Note that, the most time-consuming part is the solution of problem (3.12) with the embedded problem (3.5). If we change the embedded problem (3.5) with $\ell = L_n$, the running time drops significantly. In the case at hand, the latter amount to 53s and 51 s, respectively. The algorithm used to perform the optimization in Eq. (3.12) was Levenberg–Marquardt [36].

In Figs. 4–11 we show the recovered leverage function and the local volatility surface using the benchmark method in Section 3.2 and our proposed method shown in Section 3.3.

*Synthetic data*

We first display the results obtained by producing synthetic data. The interpolation of $L(t, S)$ is the standard interpolation of Matlab's plot noting that the inverse problem is solved in a finer mesh than the sampling of the data.

*Real data*

From Figs. 4–11, it can be seen that:

- the benchmark method is not stable and it fails to converge for large log-moneyness;
- the results of the benchmark method have more noise than the proposed method;
- the proposed method converges to the ground truth leverage function for shorter maturity than the benchmark method;

In Figs. 6, 7, 10 and 11, we show the recovered leverage function and local volatility for the two methods at different times. The proposed method and the benchmark method agree around at-the-money, but for deep in-the-money and out-of-the money log-moneyness, the corresponding local volatility curve of the benchmark method is distant from the market's local

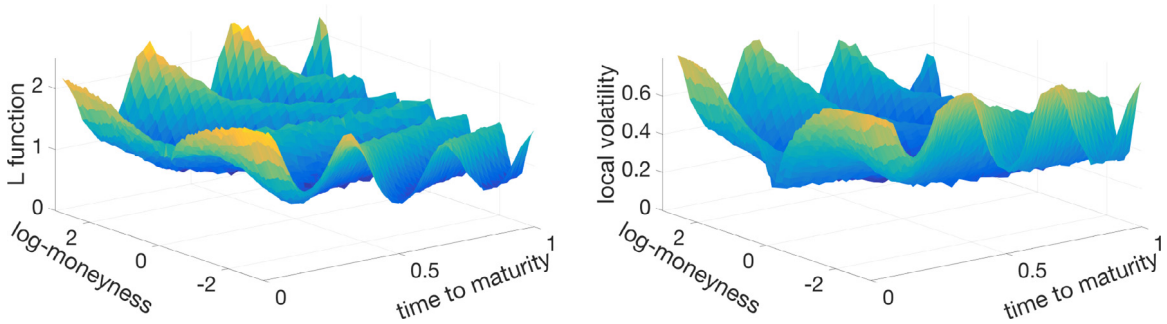



**Fig. 4.** Leverage function (left) and the local volatility surface (right) computed with the benchmark method in the synthetic data example.

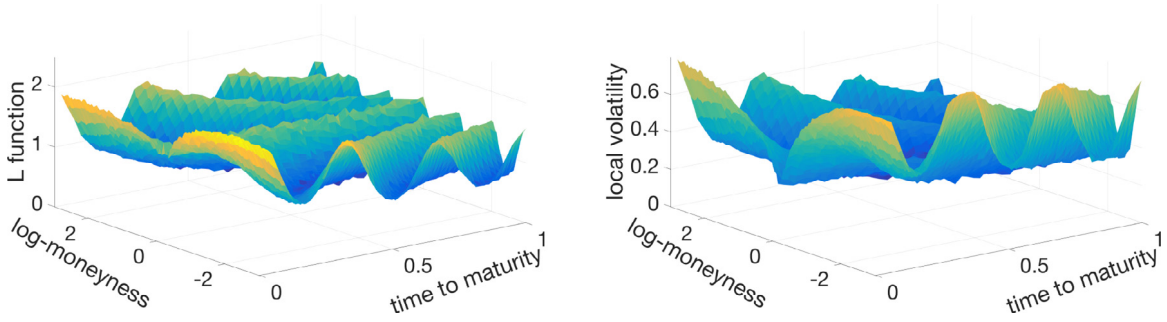



**Fig. 5.** Leverage function (left) and the local volatility surface (right) computed with our proposed method in the synthetic data example.

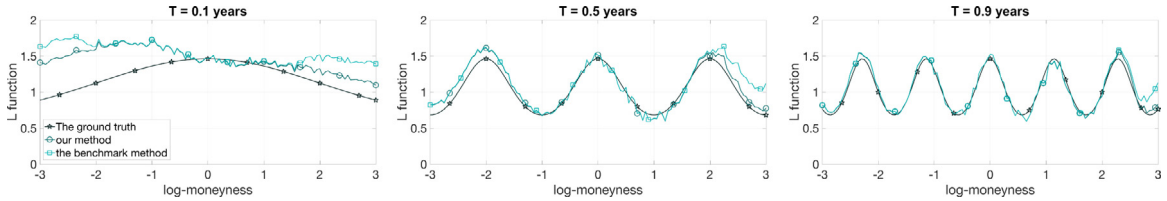



**Fig. 6.** The leverage function in the synthetic data example: the ground truth (with stars), the benchmark method (with squares) and our method (with circles).

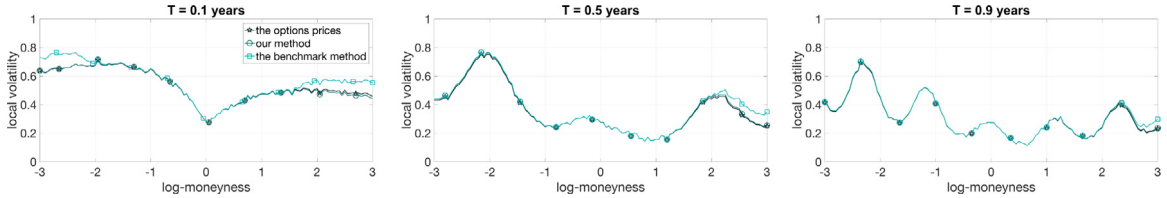



**Fig. 7.** The local volatility surface in the synthetic data example: the ground truth (with stars), the benchmark method (with squares) and our method (with circles).

volatility surface. We have shown three maturities, but this phenomenon can also be observed for all the other maturities. This phenomenon is observed more prominently in the synthetic data example, where we have considered larger log-moneyness bounds.

Once we estimate the leverage function $L$, we can recover the local volatility surface using the Alternating Direction Implicit (ADI) method with the leverage function at both times, $t_n$ and $t_{n+1}$. Comparing with the ground truth of local volatility surface in the synthetic data example, we can calculate the relative residuals. In Table 5, we present the relative residuals for the synthetic data in two intervals of the log-moneyness, which are $[-3, 3]$ and $[-2, 2]$. We also report the relative residuals of the real data example for option prices and implied volatilities. For both examples, we see that the proposed method generates better results with relative errors significantly smaller than the benchmark method. We would like to point out that this failure of convergence of the benchmark method is not related to a boundary issue. Indeed, the numerical

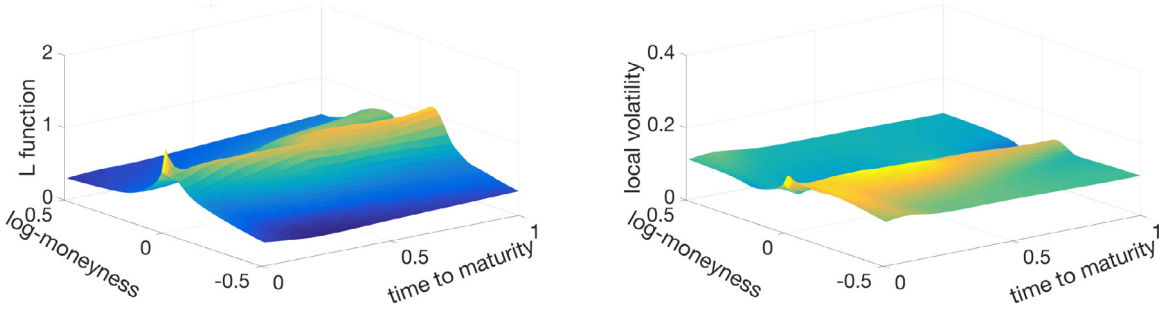



**Fig. 8.** Leverage function (left) and the local volatility surface (right) computed with the benchmark method in the real data example.

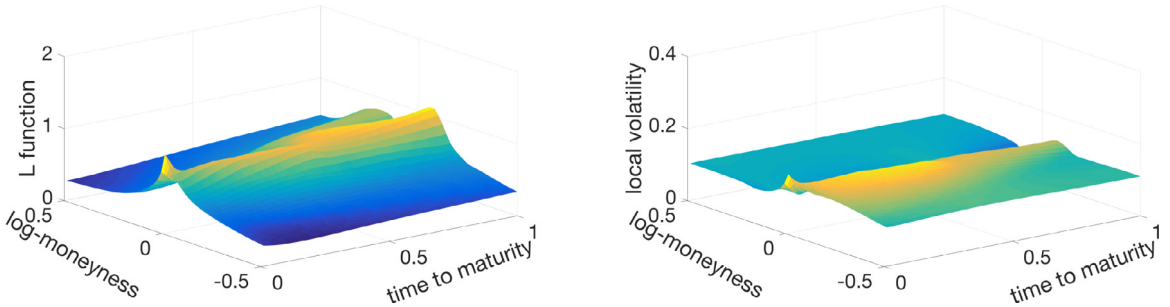



**Fig. 9.** Leverage function (left) and the local volatility surface (right) computed with our proposed method in the real data example.

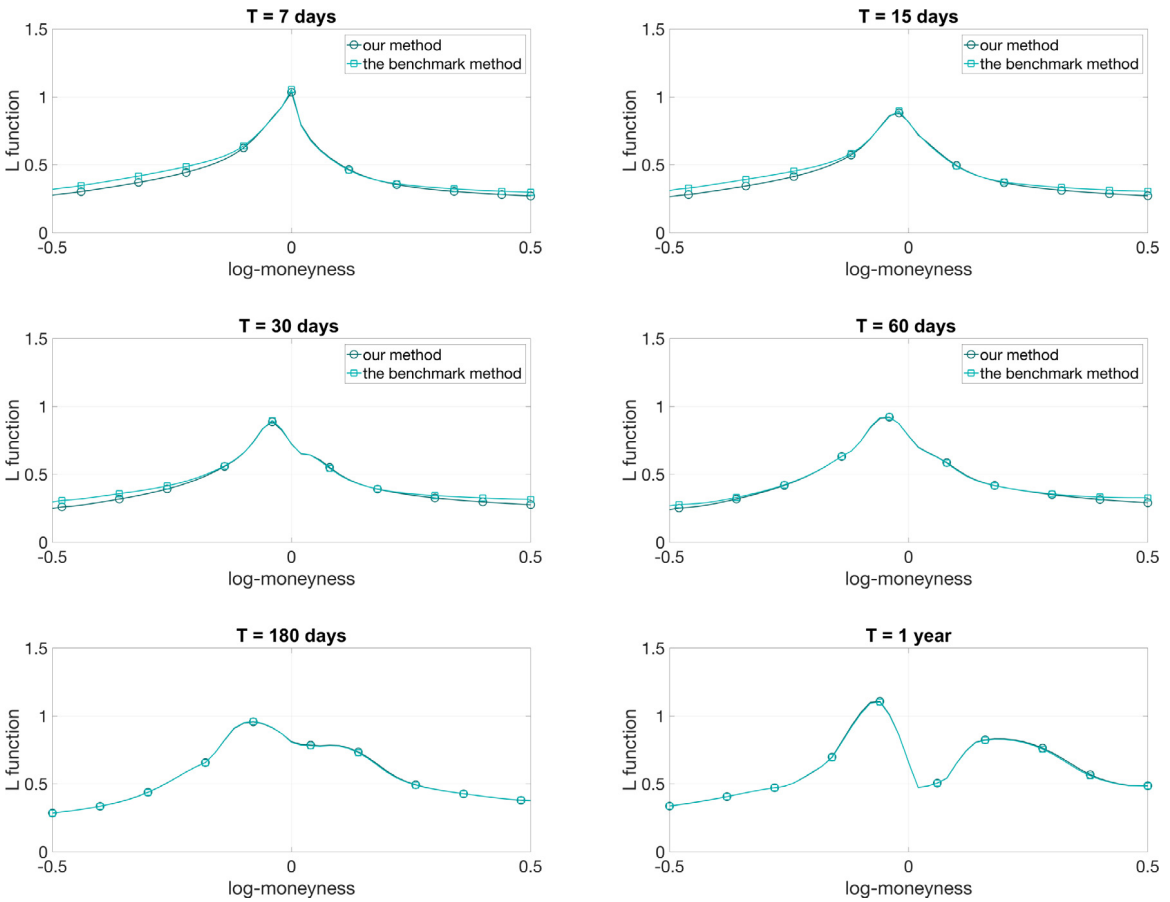



**Fig. 10.** The leverage function in the real data example: the benchmark method (with squares) and our method (with circles).

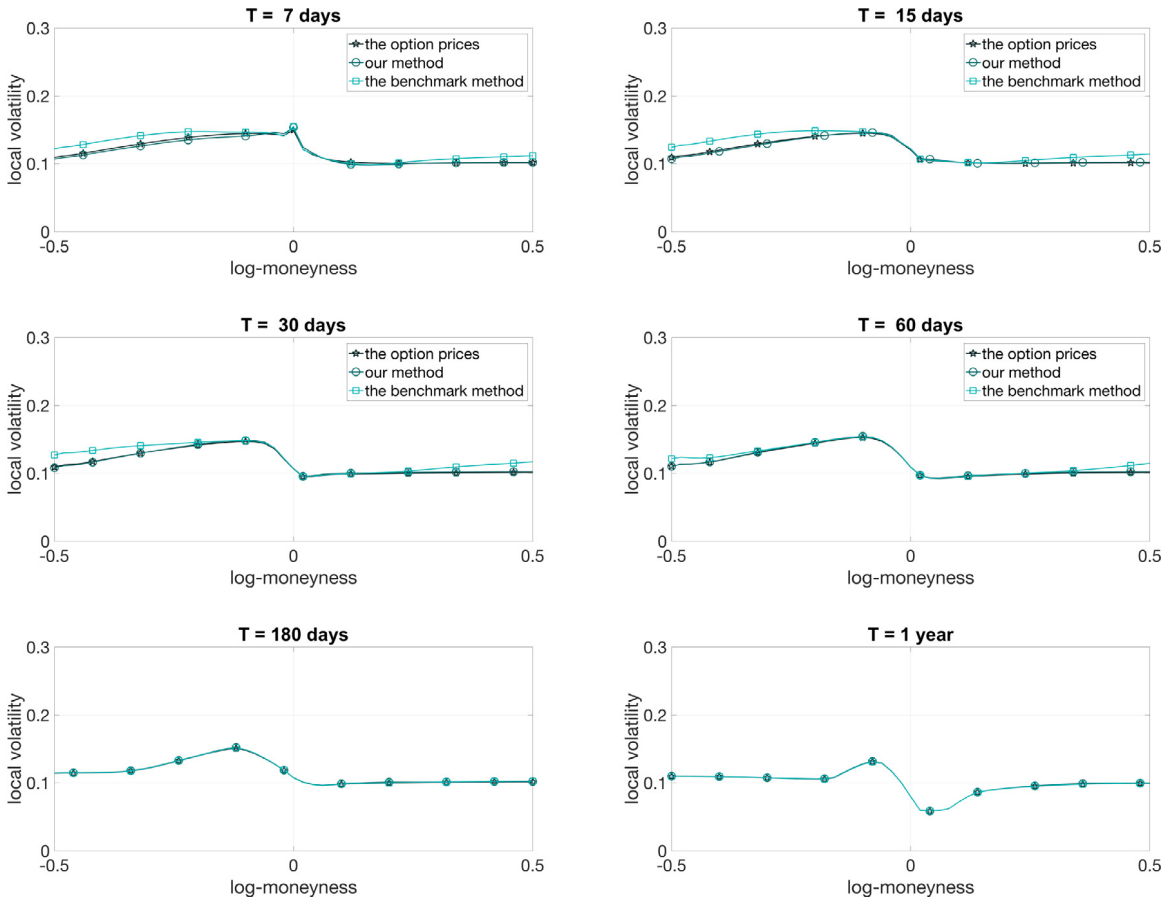



**Fig. 11.** The local volatility surface in the real data example: computed from option prices (with stars), the benchmark method (with squares) and our method (with circles).

**Table 5**
Relative residuals and running time.

| Example | Benchmark method | Proposed method |
|---|---|---|
| Synthetic in $[-3, 3]$ | 7.92% | 1.40% |
| Synthetic in $[-2, 2]$ | 2.07% | 1.09% |
| Synthetic running time | 47 s | 1.0525 h |
| Real - option prices | 2.88% | 0.73% |
| Real - implied vol | 5.21% | 3.47% |
| Real running time | 44 s | 1.129 h |

**Table 6**
The statistics of the reconstruction for the synthetic example with 100 runs.

| | Ben. method in $[-2, 2]$ | Prop. method in $[-2, 2]$ | Ben. method in $[-3, 3]$ | Prop. method in $[-3, 3]$ |
|---|---|---|---|---|
| Min. | 1.99% | 1.09% | 7.73% | 1.39% |
| Max. | 2.20% | 1.20% | 8.10% | 1.51% |
| Mean | 2.09% | 1.14% | 7.92% | 1.45% |
| Std | $4.3996e-04$ | $2.1864e-04$ | $7.6778e-04$ | $2.2818e-04$ |

experiment with real data that consider smaller log-moneyness intervals has similar results to the truncated version of the results we have found for the synthetic case.

To validate the performance of the proposed algorithm we ran a batch of 100 reconstructions for the same computed data tainted with different realizations of a pseudo-random variable. In Table 6, we summarize the statistics of the reconstructions comparing the benchmark and the proposed method.

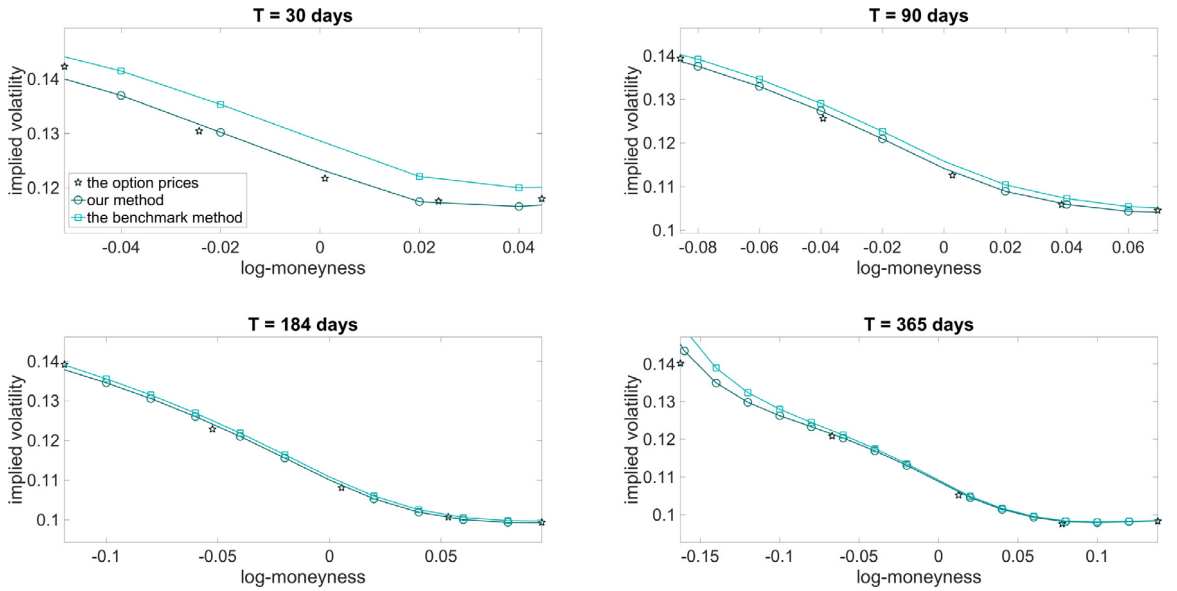



**Fig. 12.** The implied volatility surface in the real data example: computed from option prices (with stars), the benchmark method (with squares) and our method (with circles).

*The benchmark method is not exact*

One might think, at a first glance, that the benchmark is *exact* in the sense that $L^B$ gives exactly the same local volatility $\sigma_{loc}$. This would only be true in the case where we compute the numerical solution of the Fokker–Planck equation, $p^B_{n+1,i,j}$, setting $L_{n+1,i} = L^B_{n,i}$ in the numerical method chosen to solve this PDE. However, the leverage function $L^B$ does not satisfy this restriction, obviously. The correct computation of the implied and local volatilities needs to consider both values $L^B_{n,i}$ and $L^B_{n+1,i}$ for the numerical computation of $p^B_{n+1,i,j}$.

One could argue that the benchmark method, when using the misspecified numerical procedure to compute $p^B_{n+1,i,j}$, gives the exact values of the implied volatility and that our proposed method, as the reader will notice, does not. However, the computation of the price of any other financial derivative would not set the leverage function at the nodes $t_n$ and $t_{n+1}$ equal. Therefore, the correct way to compute the implied volatility would be to consider the complete knowledge of the leverage function.

As can be seen in Fig. 12, the implied volatility obtained from the benchmark method is much further away from the real implied volatility than the one obtained by our proposed method.

## 5. Conclusion and future work

We have studied the calibration of the Stochastic Local-Volatility model and proposed a numerical method based on the Tikhonov regularization framework. We compared this proposed method with a benchmark method based on PDE techniques defined in [12] under two different numerical examples. Under both cases, we have observed that the proposed method is more robust and has significantly smaller relative error when compared to the benchmark method.

The conclusion from our numerical exercises, that corroborates the theoretical reasoning, is that, when compared to the benchmark, the proposed method

- is more robust against the added noise;
- is more resilient to instabilities in the regions of low probability density of the spot price and instantaneous variance;
- approximates the implied and local volatility better at short maturities;
- respects the data in the sense that we do not apply interpolation. More precisely, the benchmark method requires the knowledge of the local-volatility on the same mesh as the one used to solve the Fokker–Planck Equation (3.2) .

Moreover, since our proposed method is aimed to improve the error created by using Eq. (3.10) to update the leverage function, we would have observed the same improvement documented in Section 4.3 if we had used the adjoint method proposed in [16] to solve the related Fokker–Planck PDE.

Future development could consider the implementation of the online calibration procedure of [30]. This could not be achieved for the benchmark method. Another avenue would be to explore the fast mean reversion stochastic volatility setting conjoined with the local volatility surface estimation as described in [37].

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## Appendix. Specification of the ADI method

To solve Eq. (3.2) numerically, we apply the finite difference Douglas–Rachford (DR) method [22]. For completeness, we shall now give the details of the implementation.

We suppose $(t, S, V) \in [t_{\min}, t_{\max}] \times [S_{\min}, S_{\max}] \times [V_{\min}, V_{\max}]$. The discretization contains $N_S + 1$ nodes in $S$ direction, $N_V + 1$ nodes in $v$ direction and $N_t + 1$ nodes in $t$ direction. By using the central difference for the first-order differentiation, all partial derivatives could be approximated as follows:

$$\begin{aligned}
\frac{\partial(Sp)}{\partial S} &\approx \frac{S_{i+1}p_{n,i+1,j} - S_{i-1}p_{n,i-1,j}}{2\Delta S} =: \frac{\delta_S^S p_{n,i,j}}{2\Delta S}\\
\frac{\partial\big((m-V)p\big)}{\partial V} &\approx \frac{(m-V_{j+1})p_{n,i,j+1} - (m-V_{j-1})p_{n,i,j-1}}{2\Delta V} =: \frac{\delta_V^{m-V} p_{n,i,j}}{2\Delta V}\\
\frac{\partial^2\big(VL^2(t_n,S)S^2p\big)}{\partial S^2} &\approx \frac{1}{(\Delta S)^2}(V_jL^2(t_n,S_{i+1})S_{i+1}^2p_{n,i+1,j}\\
&\quad - 2V_jL^2(t_n,S_i)S_i^2p_{n,i,j} + V_jL^2(t_n,S_{i-1})S_{i-1}^2p_{n,i-1,j})\\
&=: \frac{\delta_{SS}^{VL^2(t,S)S^2}p_{n,i,j}}{(\Delta S)^2}\\
\frac{\partial^2(Vp)}{\partial V^2} &\approx \frac{V_{j+1}p_{n,i,j+1} - 2V_jp_{n,i,j} + V_{j-1}p_{n,i,j-1}}{(\Delta V)^2} =: \frac{\delta_{VV}^V p_{n,i,j}}{(\Delta V)^2}\\
\frac{\partial^2\big(VL(t_n,S)Sp\big)}{\partial S\partial V} &\approx \frac{1}{4\Delta S\Delta V}(V_{j+1}L(t_n,S_{i+1})S_{i+1}p_{n,i+1,j+1} +\\
&\quad V_{j-1}L(t_n,S_{i-1})S_{i-1}p_{n,i-1,j-1}\\
&\quad - V_{j+1}L(t_n,S_{i-1})S_{i-1}p_{n,i-1,j+1} -\\
&\quad V_{j-1}L(t_n,S_{i+1})S_{i+1}p_{n,i+1,j-1}) =: \frac{\delta_{SV}^{VL(t,S)S}p_{n,i,j}}{4\Delta S\Delta V}
\end{aligned}$$

We replace the derivative in Eq. (3.2) by finite difference quotients. We then define the discretized system for the approximation $p_{n,i,j}$ for $p(t_n, S_i, V_j)$ given by the $\theta$-scheme:

$$(1 - \theta A_1 - \theta A_2)p^{(n+1)} = [1 + A_0 + (1-\theta)A_1 + (1-\theta)A_2]p^{(n)} + O(\Delta t^3)$$

for $i = 1, 2, \ldots, N_S - 1, j = 1, 2, \ldots, N_V - 1$ and $n = 1, 2, \ldots, N_t - 1$, where $p^{(n)} = \{p_{n,i,j}\}_{i,j=0}^{N_S,N_V}$ and

$$\begin{aligned}
A_0 &:= \frac{1}{4}\rho\xi R_{SV}\delta_{SV}^{VL(t,S)S},\\
A_1 &:= R_{S2}\delta_{SS}^{VL^2(t,S)S^2} + \frac{1}{2}(r-d)R_S\delta_S^S,\\
A_2 &:= \xi^2 R_{V2}\delta_{VV}^V + \frac{1}{2}\kappa R_V\delta_V^{m-V},
\end{aligned}$$

$$R_S := \frac{\Delta t}{\Delta S},\ R_V := \frac{\Delta t}{\Delta V},\ R_{S2} := \frac{\Delta t}{\Delta S^2},\ R_{V2} := \frac{\Delta t}{\Delta V^2},\ R_{SV} := \frac{\Delta t}{\Delta S\Delta V}.$$

The Douglas–Rachford method (DR method) is then defined as:

$$\begin{aligned}
(1 - \theta A_1)W &= [1 + A_0 + (1-\theta)A_1 + A_2]p^{(n)}\\
(1 - \theta A_2)p^{(n+1)} &= W - \theta A_2 p^{(n)}
\end{aligned}$$

Note that, here, for notational reasons, we assume the rate $r - d$ is constant. In our experiment, with a slight modification of $A_1$, we developed the method for the case of $r - d$ being time-dependent.

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